Customers Like You

To: Sarah Okafor, Head of E-Commerce From: Data & ML 13 October 2026

Use SVD as a candidate for a small recommendation trial.

On 1,910 returning customers with eligible new purchases, SVD placed at least one actual new purchase in the top 10 for **21.3%**, compared with **17.4%** for unseen best sellers. The difference is +3.9 percentage points. PCA reached 21.5% on the same customers.

Method and what compression loses

We used Truncated SVD with 50 components. It retains the largest singular values to preserve strong shared purchase patterns, and works directly on the sparse purchase table. Products are suggested through similarity to items a customer bought. PCA, our required comparison, describes departures from the average customer. Keeping fewer patterns loses detail, especially for uncommon products. Relative table reconstruction error is 79.0%; recommendation hit rate measures purchase matching separately.

Size and nightly speed

Factors occupy **3.67 MB**, versus 161.83 MB for the full dense table (97.7% smaller) and 5.00 MB for sparse storage. Keeping purchase history and top-10 caches brings the array total to 9.33 MB, excluding labels and file overhead. Fitting plus product-page refresh took 1.04 seconds; scoring all customers took another 1.13 seconds. These are local, single-thread timings; raw Excel loading is separate.

Customer groups for marketing

The two-component map contains 3 exploratory groups: Focused kitchen and gifts (4,554); Heart and home decor (537); Children's lunchware and gifts (391). Names reflect characteristic products. Country mixes are reported separately; groups also differ in purchase breadth. The map is a simplified view and needs campaign testing.

Risks and next steps

Test dates were 10 Oct to 09 Dec 2011. Evaluation covers 71.4% of all new test purchase pairs after customer and catalogue exclusions. New customers and unseen products need a best-seller fallback. Popularity, wholesale volume and seasonal demand may influence results. Returns were excluded; missing purchases do not express dislikes. Validate on a later period, review suggestions and run a controlled trial. Keep customer identifiers private.

Source: Chen (2012), Online Retail II, UCI. doi:10.24432/C5CG6D. CC BY 4.0. Full evidence and reproducible analysis accompany this memo.